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Scaling medical AI across clinical contexts

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Medical artificial intelligence (AI) tools, including clinical language models, vision–language models and multimodal health record models, are used to summarize clinical notes, answer questions and support decisions. Their adaptation to new populations, specialties or care settings often relies on fine-tuning, prompting or retrieval from external knowledge bases. These strategies can scale poorly and risk contextual errors—outputs that appear plausible but miss critical patient or situational information. We envision context switching as an emergent solution. Context switching adjusts model reasoning at inference, without retraining. Generative models can tailor outputs to patient biology, care setting or disease. Multimodal models can switch between notes, laboratory results, imaging and genomics, even when some data are missing or delayed. Agent models can coordinate tools and roles based on task and user context. In each case, context switching enables medical AI to adapt across specialties, populations and geographies. This approach requires advances in data design, model architectures and evaluation frameworks, and establishes a foundation for medical AI that scales to an infinite number of contexts, while remaining reliable and suited to real-world care.

Medical foundation models show strong performance on benchmarks such as note summarization and board-style question–answering, and can support specific steps in clinical decision-making[1–4]. They can streamline clinical workflows by reducing documentation burden and accelerating information synthesis[5,6], and in controlled settings they can achieve accuracy approaching expert judgment[7–9]. These successes raise expectations for clinical use beyond benchmark conditions[10], including use across populations, specialties and care settings. Outside benchmark conditions, performance can degrade sharply[11]. Real-world care involves patients with comorbidities, atypical and heterogeneous disease presentations, and rapidly evolving standards of care that are only weakly captured in training data[12,13]. In these settings, model outputs often sound correct but miss critical contextual details[14,15]. These contextual errors expose a central limitation: AI models recognize statistical patterns[16,17] but struggle to adjust their reasoning when the clinical context shifts.

This gap stems from a two-stage paradigm of large-scale pretraining followed by narrow, task-specific fine-tuning on datasets with curated labels for tasks[1,2,7] (such as triage, diagnosis or prognosis[1,2,7,18]). Performance depends on how well the training distribution matches the deployment setting. When distributions vary across hospitals, populations or specialties, model performance drops[17,19,20]. Prompting[2,21],

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retrieval or multimodal inputs[5,8] can improve results, but these strategies require repeated fine-tuning, which cannot scale across all possible clinical contexts.

As AI use expands across healthcare, context awareness becomes urgent. Models must adapt automatically to differences in users, health systems, geographies, diseases and populations[22,23]. Without such adaptation, models rely on manual adjustment, which limits scalability and introduces errors[14,15]. Precision medicine demands care plans tailored not only to the patient but also to the constraints of the health system that delivers the care[22].

We envision context switching to become the defining paradigm of medical AI. This capability adjusts outputs as clinical context shifts (Fig. 1). Context switching is not primarily about acquiring new medical knowledge; it is about using the available information to adjust model behavior at inference time (see Box 1 for brief descriptions of key context switching approaches). With context switching capabilities, a model that interprets chest X-rays in a tertiary hospital should still function in a clinic with partial data; a model trained on adult cardiology should account for pediatric physiology when used in children; and a documentation model should adjust tone and content when writing for patients versus clinicians. These adjustments occur during inference, not through retraining. Rather than operating as rigid pipelines, context switching medical AI must function as adaptable systems that respond as clinical variation arises.

Context switching builds on three pillars. First, data strategies must capture context-specific signals from patient records and medical knowledge. Second, models must adjust their reasoning at test-time to differences in data, user roles and disease settings. Third, evaluation must expose how models behave across real-world variability and identify failure modes that guide more generalizable design. These directions chart a path for medical AI that learns to reason in context rather than remaining tied to static distributions. We envision context switching as the guiding paradigm for the future of medical AI, one that aligns model behavior with the complexity of clinical practice and the diversity of patients. In the sections that follow, we discuss the early strategies for context switching in medical AI (for example, generative, multimodal and agentic models) and their limitations, and identify several concrete opportunities for further development. Additionally, we describe three clinical vignettes in which context switching is a necessity and outline case solutions using the three pillars.

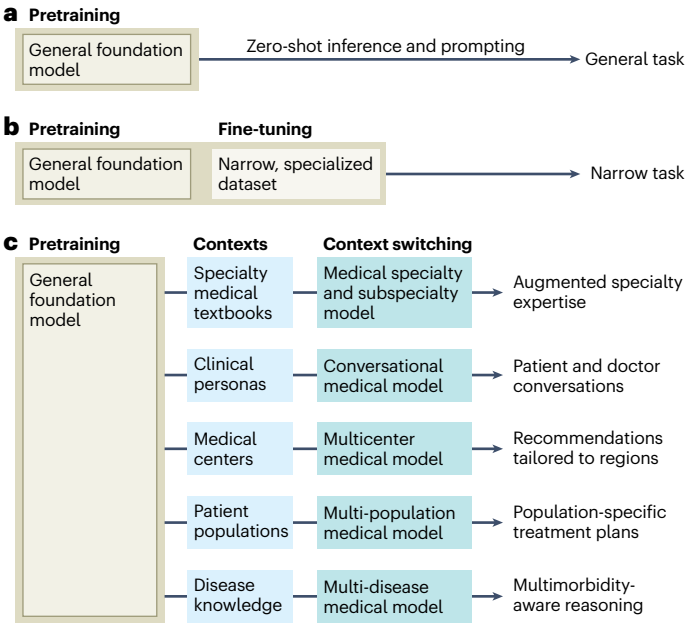

Fig. 1 | From pretraining to contextual adaptation in medical AI. a, Foundation models trained on large, general-domain data can perform a wide range of tasks but often fail to generalize in specialized medical settings. Their performance declines when applied to domains that require localized or domain-specific knowledge. **b**, Prompting, retrieval from external databases and fine-tuning can align general models with specific medical tasks. However, performance gains diminish when distribution shifts are large, such as those due to different diseases, populations or care settings, and fine-tuning is inefficient and hard to scale. **c**, Context switching models adjust in real-time to changes in specialty, disease, user role or data availability, without retraining. This shift supports broader clinical utility by reducing reliance on labeled data and serving many tasks, users and clinical settings. Context switching models adapt to an infinite range of medical specialties, healthcare roles, diseases and populations.

Why prompting and fine-tuning fall short

Prompting and fine-tuning (Box 1) are the most common strategies for adapting foundation models to medical tasks. While effective in controlled settings, these approaches struggle to support adaptation across the diversity of real-world clinical contexts. In practice, this adaptation is often implemented through natural-language interaction: foundation models act as an interface between computational systems, clinicians and patients[5,6,8,9]. For example, personalized chatbots can extend timely support to patients with rare, complex or stigmatized conditions who often face delays in access[4]. By using language as a shared interface[24], prompts can supply situational details and steer model behavior, providing a form of context adaptation. A prompt may include input–output examples that the model uses to infer the expected structure and content of the task[25–29]. When a sufficient number of examples are available, the model can be adapted through fine-tuning to improve performance on the new task[30–33].

Pretraining followed by prompting and fine-tuning represents an initial step toward context-adaptive models, but remains limited because adaptation depends on carefully curated examples and repeated fine-tuning for each new task[17,19,20,34]. In medical contexts, aligning a model's training with the intended output is difficult because clinical presentations involve subtle patterns, such as comorbidities or atypical findings, that general prompts rarely capture[12,13,35]. Prompting also depends on human–computer interaction factors, including the clinical expertise of the user, cognitive biases[36] and the time available to craft and iterate on prompts[37]. These constraints intensify in high-pressure settings, such as emergency care[37], and in resource-limited settings, such as rural clinics. Prompting frameworks that automatically optimize prompts without updating model parameters can reduce manual effort[38–40] and improve performance on medical question-answering tasks[41–44], indicating that context switching can emerge from optimized prompting and in-context learning (for example, providing sufficient context in the query) (Box 2). However, even when models perform well in isolation, they may not work effectively when used alongside clinicians[11,45]. Large language models often encode cognitive biases and display sycophantic behavior[46,47], distorting responses and unpredictably influencing the clinicians who prompt them.

Prompting a foundation model while relying solely on its parametric knowledge limits its capacity to handle distribution shifts in target labels[5,6,8,9,34], such as updates to disease classifications, evolving diagnostic criteria or newly recognized syndromes[12,13,35]. The verification of generated outputs through retrieval-augmented generation and knowledge graphs[48] can improve contextual relevance in medical question-answering tasks[49–52], indicating their potential to enable context switching in real-time settings. Context discovery via retrieval-based methods (Box 1) reduces errors due to confabulation, improves accuracy and increases efficiency, but it requires context-relevant data in the vector database[53–56]. However, such data are scarce for patients from underrepresented groups, from low-resource regions, with rare conditions or who speak less widely used languages[57]. Although effective, fine-tuning requires curated data and risks overfitting to narrow tasks while degrading performance on others

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BOX 1

Early approaches to context switching in medical AI

Several approaches illustrate how models begin to adapt their behavior in varying degrees across contexts:

Prompt engineering. Input instructions supply the missing context and narrow the search space of possible outputs[29].

In-context learning. Zero-shot or few-shot examples in the prompt show the model how to perform a task in a new setting[27,28].

Context discovery. Algorithms identify, retrieve and compress the most relevant evidence from external data sources, re-rank it by user constraints and feed it into the model[115].

Supervised fine-tuning. Models update on curated datasets to perform a target task. Parameter-efficient fine-tuning achieves similar performance with fewer resources; full fine-tuning is reserved for cases that require large behavioral shifts[26,30–33].

Post-pretraining adaptation. Multi-round refinement adjusts model behavior and aligns outputs with user intent. Techniques include instruction tuning to follow instructions[116,117], domain-adaptive and task-adaptive pretraining to capture specialized vocabulary and patterns[118], and adaptations that teach models to use tools and call functions[119]. Preference optimization, often implemented through reinforcement learning from human feedback, further aligns outputs with user expectations[99].

Agent models. Autonomous large language model agents plan, reason and act across contexts[90]. Specialized components for planning, memory, tool use or task decomposition collaborate on multistep workflows and adjust outputs based on feedback from one another.

(for example, catastrophic forgetting; Box 2). To improve generalization, methods have incorporated large volumes of unlabeled data through unsupervised or self-supervised learning. But unlabeled data alone cannot replace targeted supervision in contexts that demand reasoning over complex relationships, disambiguation of subtle clinical signals or alignment across diverse modalities. Supervised examples define task boundaries, highlight relevant features and direct the model toward clinically actionable information. Without such guidance, models may learn spurious correlations or overlook rare or inconsistently recorded phenotypes. Only models that adapt at inference can scale safely across the diversity of clinical settings.

These limitations, along with patient privacy concerns, curb the scalability of prompting and fine-tuning as well as their potential to become strategies for adapting models to real-time clinical variation[58]. Sharing large language models trained on patient data is often not possible due to concerns about privacy leakage and reidentification[59]. These restrictions prevent health systems from training models on local data and transferring them across other health systems for continued fine-tuning. As a result, models cannot be incrementally trained across diverse patient populations and care settings, which has been a longstanding practice of medical AI. Without privacy-preserving mechanisms, collaborative model development across health systems remains infeasible. By dynamically tailoring outputs to different contexts (for example, patients or health systems) at inference, context switching can enable context-aware reasoning while preserving patient privacy.

Foundation models that rely only on prompting and fine-tuning can retrieve medical facts, but they often do not adapt well when the task or clinical setting changes. In real care, clinicians apply general principles to new situations by updating their thinking as new information and constraints emerge[36,60]. Context switching aims to give models a similar ability: to adjust their outputs to the patient and care setting, not just repeat what they have seen before[61].

Steering outputs to match the clinical context

Models can adapt by changing what they generate to fit the clinical situation. This approach addresses the limitations of prompting and fine-tuning by steering outputs using signals about the patient, the care setting and the intended user. Generative models produce new content, such as patient trajectories[62,63], medical images[64] or passages of free text[65], and their outputs must match both the clinical context and the use case. In practice, this steering can be done by conditioning output generation on structured context, patient features or other contextual inputs. Across common generative model families, including flow-matching models, diffusion models and variational autoencoders, contextual inputs can steer what the model produces so it fits different clinical domains and settings[66–68]. Beyond prompt engineering and in-context learning, post-training methods can further shape model behavior, so outputs remain appropriate across clinical contexts (Box 1).

Generative models allow a model to tailor its output to the intended reader, such as a specialist, a trainee or a patient with limited health literacy[69–71]. In practice, this tailoring can be achieved by providing clear cues about the recipient and adjusting how the model writes, including the level of detail, wording and tone[72,73]. For example, a model that generates patient instructions should use different vocabulary and explanation depth for a specialist physician, a medical trainee and a patient[69,70]. When generating summaries for a care team, the model should distinguish between electronic health record (EHR) documentation, specialist referrals, and insurer justifications. Context switching supports these shifts by interpreting task-specific cues and adjusting the output structure accordingly. Domain- and task-adaptive pretraining (Box 1) can refine a model's vocabulary and patterns to reflect the intended users (for example, physician, trainee or patient) or use cases (for example, EHR documentation, medical referrals or insurance documentation). Still, models must be able to detect when errors accumulate during output generation to minimize the contextual error of the generated output (Box 2). Although first developed in general-domain language models, context switching strategies extend naturally to medical AI, where language serves as the interface between humans and machines[24].

Context switching across data modalities

Clinical decisions rarely rely on a single source of information. Clinicians weigh symptoms, laboratory results, imaging, prior notes, and guidelines, often under conditions where some data are missing, delayed, or conflicting. For medical AI to function reliably across care settings, it must similarly decide which information to use, which to ignore, and how to adjust as new data become available. Context switching across data modalities is therefore essential for scaling medical AI beyond narrow, well-controlled settings.

Multimodal models can support this need by integrating diverse data types[32,33], including structured EHRs, unstructured clinical notes, medical knowledge (for example, textbooks, research articles and guidelines), laboratory test results, imaging modalities (for example, radiology and pathology)[30,31,74,75], and genomic information[76–78]. Different model designs combine these data in different ways: some process each modality separately before integrating them, others

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BOX 2

Opportunities for context switching in medical AI

Context switching is an emerging paradigm with multiple opportunities for innovation. Five open challenges in medical AI highlight where context switching can have immediate impact:

Insufficient context in the query. When context is missing, models can struggle to narrow the search space and often reach incorrect conclusions. Current models fill these gaps from parametric knowledge. Context switching could identify the missing information and prompt the user to supply additional details, producing more accurate responses.

Static context encoding. Retrieval methods often encode external knowledge in ways that restrict extrapolation from small fragments (for example, sentences) across large sources (for example, dense documents). Dynamic, context-aware encoding could preserve nuance and prevent evidence dilution, enabling more reliable reasoning.

Catastrophic forgetting. In continual and lifelong learning, updating model parameters can improve domain-specific performance while degrading general capabilities. Analogous to continuing medical education, context switching could support efficient cycles of learning and unlearning to refine knowledge without erasing prior competence.

Reward hacking. In reinforcement learning, models may exploit loopholes to maximize rewards while failing to achieve the intended outcome, producing misaligned or harmful behavior. Context switching could design reward functions that are multifaceted, multistep and sensitive to context, reducing this risk.

Compounding error. Generative and agentic models accumulate errors across steps, leading to flawed outputs. Context switching could detect when errors exceed tolerable limits, identify the step at which failure arises and revise the reasoning process before continuing.

learn joint representations from raw inputs, and hybrid approaches combine both[79–82]. Cross-modal attention mechanisms, which let the model focus on related cues across data types (for example, linking an imaging finding to the relevant note text), help align semantically related information, while modality-specific encoders preserve details unique to each modality[30,32,74]. To cope with missing or variable inputs, models may downweight unavailable data, rely on alternative sources, or flexibly incorporate new modalities without retraining[77,78]. Using language as a shared interface, these systems can shift across modalities to integrate the information most relevant to the clinical context.

Multimodal context switching enables models to adjust their use of different data modalities dynamically at inference time based on the clinical setting and available inputs[83]. For example, a model may rely on histopathology images to classify a tumor when slides are available, but shift to lab values and clinical notes when imaging is missing. Beyond substituting one modality for another, multimodal context switching allows models to combine data types that are not co-observed during training, such as ECG waveforms and genetic variants in cardiology risk assessments. This behavior requires models to infer which modalities are most informative, detect redundancy or missingness, and reweight their reasoning in real-time. Retrieval-based context discovery (Box 1) can realize this capability by considering user-defined metrics on clinical informativeness in the re-ranking step. Preference optimization (Box 1) can allow a model to learn a reward function that aligns model outputs with human preferences based on a set of human-annotated examples (see 'reward hacking', Box 2). Further supporting this flexibility are mechanisms like attention-based modality selection, uncertainty-aware fusion, and mixture-of-modality experts that specialize based on the task and context[84–87].

Without context switching, multimodal models can be fragile when the available data change. A system built to combine pathology slides and clinical notes may work well when both are present, but may struggle when the slide is delayed, the note is sparse, or key tests are missing. Context switching aims to make these models more flexible: it helps them use whatever information is available, shift emphasis as new results arrive, and avoid overreliance on a single modality. This flexibility matches how clinical decisions are made in practice, where information is often incomplete and evolves over time. As clinical data become more varied and complex, models must be able to handle missingness and uncertainty while still producing outputs that remain reliable.

Reasoning in context through multistep analysis

Hybrid models offer a modular design for achieving context switching across tasks, modalities, and user roles. These models combine ideas from multimodal learning, where models jointly interpret diverse data types, and generative modeling, where outputs are conditioned on specific contextual signals. Compound models, such as agents (Box 1), provide a modular foundation for context switching. In these models, different subagents are trained to specialize in distinct tasks, data modalities[85,86], or clinical domains[88,89], and are orchestrated to solve problems that require context-specific adaptation[90]. For example, a dermatology case involving both imaging and text may trigger a vision agent specializing in skin lesion classification and a language agent that processes clinical notes describing symptom progression. Routing decisions are either optimized during training or dynamically inferred at test-time, based on features of the input or task prompt. This approach enables efficient specialization and reduces unnecessary computation, but relies on robust mechanisms to manage transitions between components.

Agents are goal-directed models that decompose complex tasks into subtasks, plan actions, and interact with data, tools, or users in real-time[88–90]. Agent models typically consist of multiple specialized agents, each trained to perform a distinct function (for example, information retrieval, question–answering, decision support or patient communication). Context switching in this setting refers to selecting and orchestrating the appropriate agent or submodel based on features of the task, user role, or available data. For example, an agent coordinating cardiovascular risk assessment may activate one subagent to extract patient history, another to synthesize lab results, and a third to generate a summary tailored to a clinician or patient.

Rather than relying on single agents to generalize across tasks, a multi-agent approach applies context switching by routing tasks to the most relevant experts. A supervisory or executive agent interprets input features, such as the type of clinical query, user background, or data modality, and assigns subtasks to specialized agents. Coordination can be implemented through gating functions, learned routing policies, or rule-based controllers that use structured metadata to direct workflows. For example, a pharmacovigilance workflow may use agents for adverse event detection, regulatory interpretation, and causal inference, with an executive agent selecting which experts to engage depending on whether the task is clinical trial monitoring or post-market surveillance. In this setting, context switching arises not

from each agent altering its reasoning strategy, but from routing and assigning the right expert agents to the clinical context.

Models that perform multistep analysis represent a key direction for context switching in medical AI. Reasoning models learn inference procedures that chain intermediate steps to address tasks, such as diagnosis, treatment selection, or risk prediction[91–93]. Reasoning models, often built on large language models fine-tuned with reinforcement learning or supervised traces, learn both the final output and the sequence of reasoning steps that generate it[93,94]. This abstraction makes reasoning models well-suited for context switching, since reasoning strategies can often transfer across tasks. For example, given a patient's symptoms, a reasoning model might infer a molecular diagnosis by sequentially identifying phenotype-gene associations, retrieving pathogenic variants, and ranking candidate genes. The same model could then apply a similar reasoning process to identify treatment options, retrieve drugs that target the implicated gene, evaluate drug indications, and prioritize therapies based on safety and efficacy. Context switching in reasoning models occurs not by changing the model architecture, but by adjusting the sequence and structure of inference steps based on the input prompt. In a novel context, reasoning models can facilitate multi-turn collaboration[34] with users, in which the model iteratively identifies the gaps in its knowledge or the given user instruction and prompts the user to provide additional guidance (Box 2).

A central challenge in developing reasoning models is designing reward functions that guide the model through intermediate steps of clinical reasoning[94]. These functions must assess the final prediction as well as the clinical relevance and plausibility of each step. For example, in differential diagnosis from patient symptoms[89], a reward function might penalize inconsistencies between symptoms and candidate diagnoses, reward logical progression in differential diagnosis from common to rare conditions, and prioritize hypotheses that fit the patient's demographic and clinical context. This structure encourages models to revise reasoning when context shifts, such as when new symptoms appear or the task changes from screening to treatment planning[95]. Reward functions must align with clinical objectives, including reducing diagnostic error, avoiding unnecessary procedures, and improving continuity of care (Box 2). Reasoning models that incorporate such context-aware rewards can adjust how they interpret and prioritize information and adapt to clinical goals and constraints.

Medical AI adapts to who, where, what it serves

Medical AI must do more than process multimodal inputs or scale across computing infrastructure. It must adapt to the contexts of care, aligning with patient needs, care settings, and clinical workflows. Variation across four key domains—specialty, geography, population and access—expose the limits of static models and make context switching a requirement for medical AI. Below, we use clinical vignettes to illustrate the importance of context switching across each of these domains.

Bridging medical specialties and disease contexts

Patients often face fragmented care in siloed health systems or resource-limited regions, leading to diagnostic odysseys and delays or errors in care[96,97]. Context switching models enable clinicians to draw on knowledge across specialties and disease domains, including areas beyond their own expertise, to guide clinical decisions (Fig. 2a).

Clinical vignette. Consider a patient with a rare disease who presents with neurological and pulmonary symptoms. The patient is referred to a neurologist and a pulmonologist, each with deep expertise shaped by shared foundational training and system-level reasoning. Yet even broad training can be constrained by framing effects that narrow the differential diagnosis to more common or organ-specific causes. If the symptoms reflect a multisystem disorder, this framing may lead to missed connections or delayed diagnosis. Context switching models can support clinicians by integrating knowledge across specialties, identifying shared pathophysiology and generating a unified diagnosis and treatment plan. A related challenge arises in polypharmacy, when multiple physicians prescribe treatments for different conditions without visibility into overlapping risks or drug–drug interactions. Context switching models can mitigate these issues by synthesizing cross-specialty knowledge, reasoning across comorbidities and tailoring treatment to minimize therapeutic conflicts.

Context switching across specialties and disease domains remains difficult because both care and model development often follow specialty-specific boundaries. Foundation models trained on siloed datasets or optimized for narrow objectives can reproduce the same limitations seen in human reasoning. Overcoming this requires models that perform multimodal reasoning and generalize across specialties, diseases and combinations of data types, even when those combinations are absent during training.

Data. Without context-specific datasets, models cannot generalize across specialties and diseases. Domain expertise can be used to curate diverse instructions (for example, chain-of-thought[98] reasoning steps that improve generalization to complex tasks) or response-preference sets (for example, human-annotated rankings of model outputs that align responses with user preferences[99]) for each specialty or disease. A useful resource for enabling context switching is a prompt library: a structured collection of multimodal prompts linked to specific specialties and disease scenarios.

Model. Medical specialties rely on complementary data for diagnosis and treatment. For example, evaluating heart failure may require electrocardiograms, blood tests and chest X-rays. In neurology, diagnosing a seizure disorder may involve electroencephalograms, brain imaging and genetic testing. Multimodal models that identify and prioritize the most relevant data for a given clinical context, such as through context discovery (Box 1), can support context switching across specialties and diseases.

Evaluation. Model generalizability is critical for context switching. A model should not fail when faced with a new context that requires a novel combination of data modalities. This necessitates benchmarks that assess whether models can identify the correct contexts, generate responses relevant to the specialty or disease, and align outputs with specialist preferences[100–103]. These benchmarks must also minimize data contamination (for example, overlap between pretraining and test datasets) to provide a valid measure of model generalizability[104,105].

Tailoring care to geography and population

Understanding the incidence, prevalence and mortality rates of health conditions in specific populations is essential for tailoring clinical decisions[106]. For example, isoniazid preventive therapy is recommended by the World Health Organization for people living with HIV in high tuberculosis burden settings, even in the absence of active tuberculosis. However, there is ongoing debate about when to initiate isoniazid preventive therapy, especially relevant at the time of HIV diagnosis, due to concerns about adherence, resistance and regional variation in tuberculosis risk[107]. These decisions depend on global guidelines as well as on local epidemiology, healthcare infrastructure and social context. For contextualized care, models need to interpret and apply guidelines in light of regional disease patterns, resource availability and population needs[106,108,109] (Fig. 2b).

Clinical vignette. Consider a patient who is susceptible to a disease that can lead to organ dysfunction. To monitor the patient's disease risk, a clinician leverages an international scoring function to regularly assess organ or tissue function using the patient's EHR data. A key challenge of designing such a scoring function is to avoid introducing contextual

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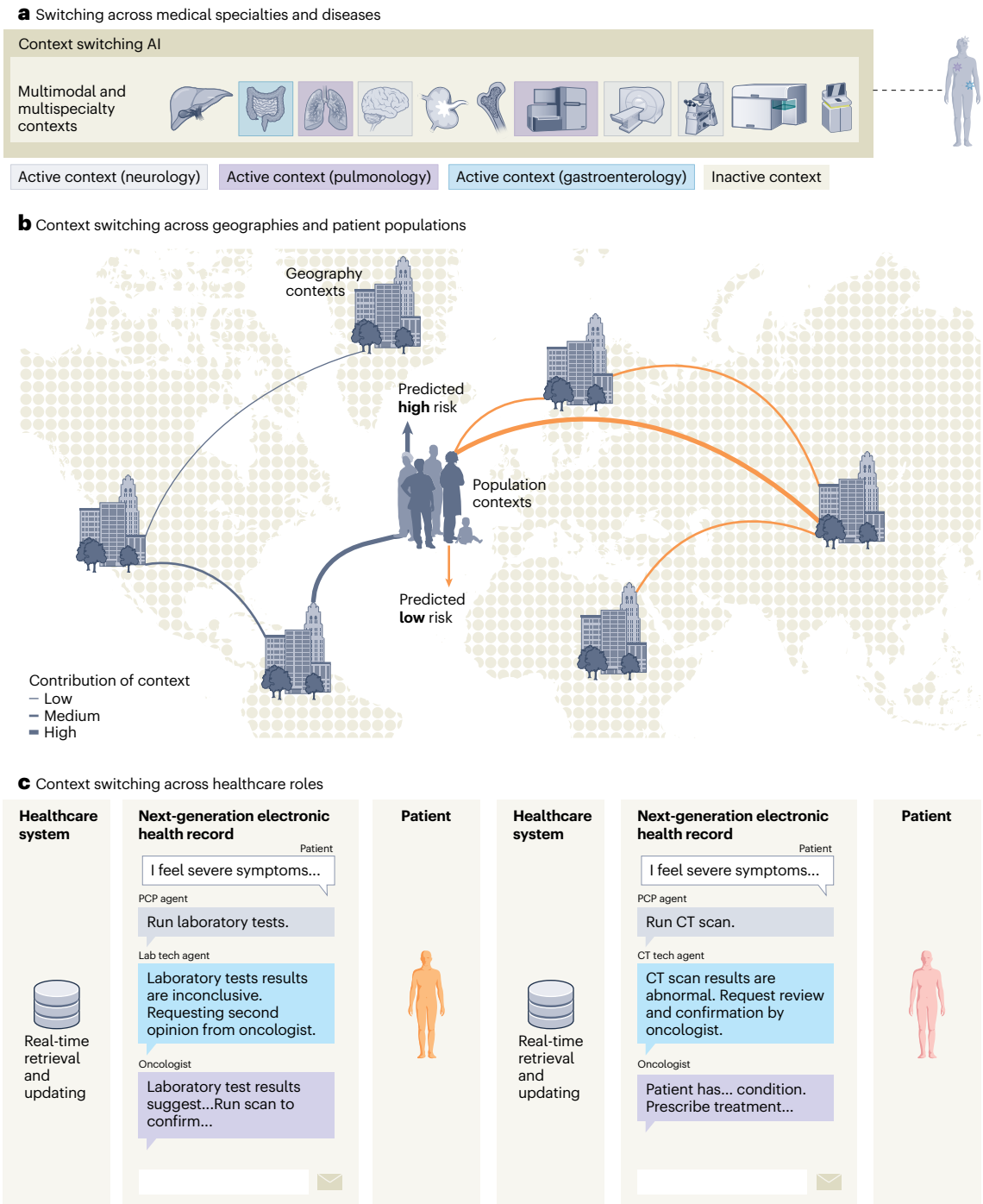

Fig. 2 | Context switching across medical specialties and diseases, geographies and populations, and healthcare roles. a, A context switching model for medical specialties and diseases can be a multimodal generative model that identifies and integrates the most relevant data modalities (for example, histopathology images, computed tomography (CT) scans and whole-genome sequencing) and clinical specialties (for example, neurology and pulmonology) to guide diagnosis and treatment. **b**, Adapting to geographic- and population-specific needs can involve a generative reasoning model capable of multistep inference, capturing both high-level reasoning (for example, shared disease risk factors) and fine-grained variation (for example, subpopulation-specific modifiers). **c**, A generative multi-agent system, where each agent represents a distinct healthcare role (for example, nurse, primary care physician or specialist), can flexibly engage the appropriate expert(s) for a given case, involve the patient throughout the decision-making process, and continuously update outputs using real-time access to the patient's EHRs.

error by assigning different scores to patients with similar clinical profiles based only on geography or demographic attributes, such as age, sex or race. These discrepancies often arise from uncalibrated models that ignore variation in healthcare delivery, disease prevalence or access to care across settings. As a result, identical physiological inputs may yield different recommendations depending on context, leading to inappropriate care. Context switching models address this by incorporating region-specific information, such as care practices, health infrastructure or disease incidence, directly into score computation. A multicenter model can switch across country-level contexts to align with local systems and guidelines, while a multi-population model can detect subgroup-specific patterns and adapt outputs accordingly.

For example, the model might recommend inpatient rather than outpatient treatment when follow-up care is difficult to access.

Data. Patient-level data vary substantially across settings, even within the same country. Differences in documentation, diagnostic workflows and data completeness can limit the portability of models. Although regional systems and guidelines also influence care, the most immediate challenge is variability in the underlying data. To support context switching, models must be trained on datasets that capture this heterogeneity. Engaging local experts and building regionally representative datasets enable models to capture both how care is delivered and why it varies across contexts.

Model. To generate context-specific recommendations, a reasoning model can integrate multistep logic that captures both general clinical objectives and local constraints. This includes treatment-level decisions (for example, choosing a therapy based on symptoms) and fine-grained adjustments (for example, adapting dosing schedules or care settings to local resource availability). For instance, a model can be trained on human-annotated chain-of-thought reasoning traces to capture this nuanced multistep decision-making process. A context switching model can identify which factors vary across populations and incorporate them into its reasoning process to produce context-appropriate outputs.

Evaluation. Working with local and regional experts makes it possible to define metrics that evaluate whether model outputs align with guidelines and common practices in specific populations and geographies. Accuracy and robustness can also be measured by the trust of the target population in the model's recommendations and their acceptance of its use in the community[110].

Democratizing access without reinforcing inequities

Access to medical care can be constrained by socioeconomic, geographic, financial and cultural factors[111,112] (Fig. 2c). Context switching can adapt to patient needs, care settings and communication styles[113], but if models are built on biased data or without input from diverse communities, they risk reinforcing inequities. The impact of context switching depends on the model's design, who develops it and whose context it reflects[114].

Clinical vignette. Consider a patient who arrives at the emergency department experiencing severe symptoms. The patient has frequently missed wellness visits. Although referred to oncology, the patient has not followed up. A typical response might be for the clinician to remind the patient to schedule the appointment, but this overlooks key contextual barriers. The patient may live far from a specialty clinic, lack reliable transportation or childcare, or be unable to take time off work. Without accounting for these constraints, the system risks recommending care that the patient cannot access: a contextual error.

Contextualized care incorporates these constraints. A more appropriate intervention can be to assign the patient a personalized medical chatbot for symptom monitoring (for example, automatic prompting by the chatbot to record symptoms or synchronizing measurements captured by mobile health monitoring applications) and real-time feedback at home. The chatbot should communicate securely with the clinical team, augment documentation and improve continuity of care. A context switching conversational model can combine clinical knowledge with the communication styles and decision roles of different healthcare professionals to engage the patient effectively and collect relevant information. By adapting in this way, context switching models can expand access to medical expertise and provide continuous, reliable points of contact between patients and their care teams.

Data. With consent, researchers can curate conversations between patients and healthcare professionals, including nurses, primary care physicians and specialists, as well as communication between clinicians. These conversations should reflect diverse populations, languages and care settings to capture variation in both patient expression and expert communication styles. Additionally, human-annotated rankings of model outputs can ensure the alignment of generated responses with the healthcare role. Relevant clinical tools and databases can be integrated to improve the accuracy and adherence to clinical guidelines of the model.

Model. A context switching conversational model can be structured as a generative multi-agent system, where each agent embodies a distinct clinical persona. Agents can be selectively activated depending on the task or patient need; for example, a nurse practitioner for symptom triage or a specialist for treatment counseling. These agents retrieve and incorporate up-to-date patient records to adjust dialogue and reasoning in real-time, based on evolving clinical context. Patients can initiate conversations with any agent or be directed to the appropriate one when uncertain. In collaboration with the intended users, guardrails and recovery policies can be implemented to maintain long-term consistency, reduce failure modes of subagents and mitigate security and data governance violations.

Evaluation. Assessing whether these models democratize care requires metrics developed alongside medical and public health experts. One metric can be the rate of successful referrals to the appropriate specialists, which reflects how well the model understands clinical roles. Appointment follow-through can serve as a proxy for patient engagement and perceived accessibility. Other relevant metrics can include symptom resolution timelines, quality of documentation and patient–clinician satisfaction.

Conclusion

We envision context switching as the defining paradigm of medical AI. Rather than building separate models for each specialty or health system, future models can adjust their reasoning at test-time to multimodal data, clinicians, patients and decisions required in context. This shift moves medical AI beyond narrow, static tools toward models that reflect the complexity and variability of clinical practice. Medical care spans diverse specialties, populations and workflows that differ across settings and evolve over time. To remain reliable, models can respond to this variation without manual adjustment. Context switching models can adapt in real-time, reduce dependence on curated data and support consistent performance across health systems. Progress must move beyond computational benchmarking toward contextual, decision-centered evaluation that aligns medical AI models with goals that matter in specific clinical contexts. Context switching AI models can scale to match the complexity of real-world clinical practice and the diversity of patients.

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All authors contributed to the writing of the paper and provided critical feedback. M.M.L. and M.Z. conceived the study and were responsible for overall direction and planning.

Competing interests

The authors declare no competing interests.

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